Hospital Bed Scheduling System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

Efficient management of hospital beds and operating rooms (ORs) is one of the most critical challenges in modern healthcare, as hospitals must handle a large number of patients while making the best use of limited medical resources. Manual scheduling methods often lead to delays, idle operating rooms, prolonged patient waiting times, and inefficient resource utilization. This project proposes a **Hospital Bed and Operating Room Scheduling System** that intelligently assigns patients to available beds and operating rooms based on multiple constraints such as patient priority, treatment duration, surgery schedules, staff availability, room capacity, and emergency cases.

The proposed system uses scheduling and optimization techniques to automatically generate the best possible allocation of beds and operating rooms while minimizing patient waiting time, operating room idle time, and overall treatment completion time. It also ensures that scheduling conflicts are avoided and that all hospital constraints are satisfied. The system dynamically updates schedules whenever emergencies arise or when patients are discharged earlier or later than expected.

The application provides separate modules for patient registration, bed management, operating room scheduling, doctor and staff availability, emergency patient handling, schedule optimization, and reporting. Hospital administrators can easily monitor bed occupancy, operating room utilization, patient status, and staff schedules through an interactive dashboard. The system also generates detailed reports and analytics that help hospital management improve operational efficiency and decision-making.

By automating the scheduling process, the proposed system enhances resource utilization, reduces administrative workload, improves patient satisfaction, and supports better healthcare delivery. The project demonstrates how optimization algorithms and intelligent scheduling techniques can significantly improve hospital operations while ensuring timely treatment, efficient use of medical facilities, and high-quality patient care.

Signature of the Guide